

Assignment Workflow Submission Guide

This guide explains how to prepare and publish a GitHub assignment by forking the provided repository, cloning the fork, committing the completed work, creating a tag, and pushing everything to GitHub.

Prerequisites

Before starting, make sure that:

- You are signed in to your GitHub account.
- Git is installed on your computer.
- You have access to the original course repository: [iomico-public/zephyr-course](https://github.com/iomico-public/zephyr-course).
- Your assignment work and any required tests are complete.
- You have configured one of the following GitHub authentication methods before cloning or pushing:
 - **HTTPS with a personal access token (PAT):** Follow GitHub's [Managing your personal access tokens](#) tutorial. The token must have permission to read and write repository contents.
 - **SSH:** Follow GitHub's [Connecting to GitHub with SSH](#) guide. It covers [generating an SSH key](#), [adding the key to your GitHub account](#), and [testing the SSH connection](#).

You only need one authentication method. Use the same method when cloning and pushing.

1. Fork the Repository

1. Open the original course repository:  [GitHub - iomico-public/zephyr-course](https://github.com/iomico-public/zephyr-course) .
2. Select **Fork** in the upper-right area of the GitHub page.
3. Confirm that your GitHub account is selected as the owner.
4. Keep the repository name required by the assignment.
5. Select **Create fork**.
6. Wait for GitHub to create the repository under your account.

The new repository is your fork. The remaining steps must use this fork rather than the original repository.

2. Copy the Fork URL

1. Open your fork on GitHub.
2. Select the green **Code** button.
3. Select the protocol that matches the authentication method you configured:

- Select **HTTPS** if you will authenticate with a personal access token.
- Select **SSH** if you configured an SSH key in your GitHub account.

4. Copy the URL shown for the selected protocol.

The HTTPS URL will have this format:

```
1 | https://github.com/YOUR-USERNAME/zephyr-course.git
```

The SSH URL will have this format:

```
1 | git@github.com:YOUR-USERNAME/zephyr-course.git
```

Replace `YOUR-USERNAME` with your GitHub username. Make sure the URL belongs to your fork and not to the original `iomico-public` repository.

3. Clone the Fork

Open a terminal and run `git clone` with the URL copied from your fork.

For HTTPS with a PAT:

```
1 | git clone https://github.com/YOUR-USERNAME/zephyr-course.git
```

For SSH:

```
1 | git clone git@github.com:YOUR-USERNAME/zephyr-course.git
```

Wait for Git to finish downloading the repository.

4. Enter the Repository Directory

Move into the cloned repository:

```
1 | cd zephyr-course
```

In the video, an empty file represents the completed assignment work. In a real assignment, complete the requested implementation and run the required tests before continuing.

5. Review the Changes

Check the current state of the repository:

```
1 | git status
```

Review the output and confirm that Git lists the files you intend to submit.

6. Stage the Assignment Files

Stage the completed assignment work:

```
1 | git add .
```

The period stages all current changes in the repository. If you only want to stage a specific file, use its path instead:

```
1 | git add path/to/file
```

7. Create the Commit

Create a commit with a short and meaningful message describing the completed work:

```
1 | git commit -m "Complete assignment work"
```

Git will display a confirmation containing the new commit and the files included in it.

8. Confirm the Working Tree Is Clean

Run `git status` again:

```
1 | git status
```

Confirm that the working tree is clean and that no required assignment files remain uncommitted.

9. Create the Assignment Tag

Create an annotated tag using the exact tag name and format specified in the assignment instructions:

```
1 | git tag -a TAG-NAME -m "Assignment tag"
```

For example, the video uses a tag similar to:

```
1 | git tag -a 11-task1 -m "assignment 1"
```

Do not replace the required assignment tag with a different name. The remote tag must match the format expected by the course.

10. Push the Commit

Push the commit to the `main` branch of your fork:

```
1 | git push -u origin main
```

The `-u` option connects your local `main` branch to the remote branch, allowing future pushes to use a shorter `git push` command.

Authentication

HTTPS with a PAT

When Git requests HTTPS credentials:

1. Enter your GitHub username.
2. Use a personal access token as the password.

Do not use your normal GitHub account password. The personal access token must have permission to modify the contents of your fork. Keep the token secure and never include it in commands, source files, screenshots, or commits.

SSH

If your SSH key is correctly configured and loaded, Git uses it automatically and does not request a username or personal access token. You can test the connection before pushing:

```
1 | ssh -T git@github.com
```

GitHub should confirm that you authenticated successfully. If the test fails, review GitHub's [SSH connection troubleshooting documentation](#).

11. Push the Assignment Tag

Pushing the branch does not automatically publish the tag. Push the assignment tag separately:

```
1 | git push origin TAG-NAME
```

Using the example tag from the video:

```
1 | git push origin l1-task1
```

Why are two push commands used? Pushing the tag transfers the tagged commit to GitHub, but it only creates or updates the remote tag reference. It does not update the remote `main` branch. Run both commands so that the commit appears on `main` and the assignment tag is also available in the remote repository.

Alternative: Push the Branch and Annotated Tag Together

Because this workflow creates an annotated tag associated with a commit on `main`, you can push the branch and its annotated tag with one command:

```
1 | git push origin main --follow-tags
```

This command can replace both `git push -u origin main` and `git push origin TAG-NAME`. The `--follow-tags` option only includes annotated tags that point to commits being pushed; it does not push unrelated tags or lightweight tags.

If you use HTTPS and Git requests credentials again, enter your GitHub username and use the same personal access token as the password. With a correctly configured SSH remote, authentication uses your SSH key automatically.

12. Validate the Remote Repository

1. Return to your fork in the web browser.
2. Refresh the GitHub page.
3. Confirm that the new commit appears in the repository.
4. Confirm that the required assignment tag is available in the remote repository.

The workflow is complete when both the commit and the required tag are visible in your GitHub fork.

Command Summary

Choose either the HTTPS or SSH clone command, replace `YOUR-USERNAME` , and use the tag name required by the assignment:

```
1 # HTTPS with a PAT
2 git clone https://github.com/YOUR-USERNAME/zephyr-course.git
3
4 # Or SSH
5 git clone git@github.com:YOUR-USERNAME/zephyr-course.git
6
7 cd zephyr-course
8 git status
9 git add .
10 git commit -m "Complete assignment work"
11 git status
12 git tag -a TAG-NAME -m "Assignment tag"
13 # Update the remote main branch
14 git push -u origin main
15 # Publish the tag; this does not update the remote main branch
16 git push origin TAG-NAME
17
18 # Alternative: update main and publish its annotated tags together
19 git push origin main --follow-tags
```